

On languages in the tactile modality

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Abstract. While linguistic research on sign languages is getting larger and larger impact in the linguistic community, much less is known about languages in the tactile modality, namely the languages used by Deaf-Blind signers. These languages are derived from their visual counterpart, although after that, they witness independent evolution. In this work, we present a detailed overview of the properties of tactile languages showing their degree of autonomy from visual sign languages and how these properties allow to re-assess some foundational aspects of human language, including the role of native users in driving linguistic change, and the impact of iconicity in transmitting meaning. The evidence discussed in the paper will give the tools to provide an informed answer to the question whether languages in the tactile-gestural modality should be considered as an autonomous externalization of human language.

Keywords: Tactile sign languages, Pro-Tactile languages, linguistic innovation, cross-modal comparison, language status

1. Introduction

Language is probably the most astonishing product of human cognition, independently from how it evolved and whether its main features are unique to mankind or shared with other species (Schlenker et al. 2022). One of its often-neglected characteristics is its resilience. That resilience is one of the key features of human language can be seen from different angles, one is certainly acquisitional: in normal conditions, language acquisition is inevitable. Hearing and deaf infants exposed to an accessible language, spoken for hearing and signed for deaf individuals, master it with very little input and following the very same milestones (i.a., Petitto and Marentette 1991, Baker and Woll 2008). Another angle where resiliency is observed is provided by those languages that do not emerge in typical environments. This is the case of the languages used by Deaf and Deaf-Blind people, namely visual and tactile sign languages: independently from how they are acquired, these languages show how humans are able to (re-)configure the externalization production and perceptual systems so that sensory-deprivations are by-passed, and language is effectively used in all its purposes. It is this second angle of resilience that will be explored in this work.

While they start as an adaptation of visual sign languages to the tactile modality, these languages soon show features of autonomous evolution. This fact is further amplified in more structured communities like the Pro-Tactile one in the US. It is then legitimate to ask whether these systems should be treated as independent and separate languages, or whether besides modality very little changes. This paper is a preliminary attempt to offer a large-perspective view on the issue and to provide the scientific bases to formulate a reasonable answer.

After defining the main properties of tactile and pro-tactile sign languages in comparison with vocal and visual languages (Section 2), we present a detailed overview of how tactile languages are more and more diverging from their visual correspondents (Section 3). In this section, we also present a novel and independent argument based on tactile iconicity that iconicity itself is not an ancillary, but rather a resilient property of human language, well integrated in the grammatical system. The composition of the community of users is introduced in Section 4, providing some important sociolinguistic elements that will be used to address the discussion on status of languages in the tactile-gestural modality (Section 5). Finally, Section 6 concludes the paper with some remarks for future research.

2. Languages in the tactile-gestural modality

The field of general linguistics is now becoming more and more acquainted to sign languages. These are natural languages emerging in Deaf communities of various sizes, environments, and various ratios of hearing/deaf users (i.a., Nyst 2012). Much less is known about languages in the tactile-gestural modality.

In this work, we use the term *visual sign language(s)* to refer to the languages used by sighted Deaf signers, but also by hearing people, like CODAs (Children of Deaf Adults) individuals who are born in signing families, parents of deaf children who decided to learn the sign language used in the local community, interpreters, teachers working in bimodal bilingual programs, students in university degrees, academics working on sign language, and people who are just interested in learning a sign language as a second language. The term *tactile sign language* is used to refer the sign language used to communicate to Deaf-Blind people, while *pro-tactile sign language* is used for a particular evolution of tactile sign languages, in which both linguistic and social innovation occurred.¹

¹ Visual language and tactile language will be used as synonymous of visual sign language and tactile sign language in the paper. It is important to keep in mind that the definition of visual language does not include written

Although sharing the same architectural design, visual sign languages radically differ from spoken languages in the way they are externalized (Sandler and Lillo-Martin 2006, Backer et al. 2016, Quer et al. 2017). Indeed, visual sign languages use the gestural apparatus in production and the visual apparatus in perception, while spoken languages use the vocal apparatus in production and the acoustic apparatus in perception. To be more precise, the production system of sign languages includes manual articulators (hands and arms) and non-manual articulators (upper part of the body, and facial expressions). In visual sign languages, information is packaged in a multi-layered fashion (Wilbur 2000, Sandler 2012) with certain properties expressed by the manual articulators, like the encyclopedic content of the lexicon and part of the functional lexicon (e.g., pronouns, determiners, and, in some cases, adpositions, question particles, and role-shift introducers); while non-manual articulators express grammatical and prosodic information, like degree morphology, the distinction between clause types (declarative vs. polar interrogative), and role-shift. One key consequence for this different externalization is that gradient and iconic aspects of reality can be more easily encoded linguistically, so in addition to conventionality and categorical contrasts, sign language also exploits gradient and non-arbitrary constructions to express meaning (Emmorey and Herzig 2003, Geraci 2009, Liddell 2003, Taub 2002).

Tactile sign languages introduce a major innovation in the way sign languages can be expressed. Differently from visual sign languages that use the visual apparatus for perception and the gestural apparatus for production, tactile sign languages use the tactile apparatus for perception and the gestural one for production. Tactile languages started as an adaptation from visual sign languages due to the communication needs of Deaf-Blind signers, but shortly developed their own features (Willoughby et al. 2018). Figure (1), shows how tactile communication works between two Deaf-Blind signers of Tactile and Pro-Tactile French Sign Language.

languages and that definition of tactile language does not include the Braille system. We also write *Deaf* and *Deaf-Blind* with capital “D” and “B” to indicate members of the Deaf and Deaf-Blind communities of visual and tactile sign language, respectively. Lower case “d” and “b” in *deaf* and *deaf-blind* indicate sensory deprivation with no commitment of the use of a signed language.

Fig.1a: Dialogue position.



Fig.1b: Monologue position



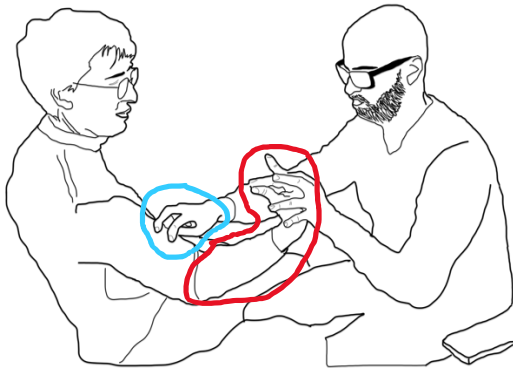
Figure (1a) illustrates the dialogue position (illustration video [here](#)), which is the position used by fluent Deaf-Blind signers in typical communication exchanges. In this setting, signer and addressee are in front of each other and the addressee puts both hands-on top of the signer who uses their hands to articulate the message. When the signer produces one-handed signs, the dominant hand is used in production (e.g., the right hand of the signer on the left in figure 1a), while the addressee perceives the message with the non-dominant hand and can use the dominant one to provide feedback that the communication flow is successful (e.g., a repeated tap on the thigh of the signer). In the monologue position (figure 1b), the participants are placed side by side and one hand only is used to produce and receive the message. This setting is used when spatial constraints do not allow the dialogue position and the communication flow is unidirectional, like during a presentation in a conference room. Both the dialogue and monologue positions have been documented for several languages including, Tactile- American (Petronio and Dively 2006), Australian (Willoughby et al. 2020), French (Schwartz 2009), Italian (Checchetto et al. 2018) and Swedish (Mesch 2001) Sign Language. When interacting with sighted people, the tactile modality is used for perception only, while the visual sign language is used in production by the Deaf-Blind signer.

As we will discuss in more detail in Section 4, the main users of tactile sign languages are deaf-blind with a variety of degrees and types of sight loss, resulting in a non-homogeneous group of signers (Mesch 2001; Willoughby et al. 2014). This aspect may lead to a variety of strategies to cope with partial or total visual loss (see also Section 3). For instance, visual residue may allow some signers to perceive at least part of the non-manual components (Willoughby et al. 2018), while in other signers this is not possible and other strategies are used. This is the case documented in Checchetto et al. (2018) for Tactile LIS, where the lack of access to the grammatical information provided by eyebrow position, an essential prosodic feature in determining sentence typing, made Italian Deaf-Blind signers grammaticalize the wh-sign

WHAT to a generic question marker, used in both polar and (redundantly) content questions, as we shortly see.

Pro-Tactile Sign Languages are an evolution of tactile sign languages to respond to the communicative needs of Deaf-Blind people and their desire of empowerment (Granda and Nuccio 2018). Originated from tactile American Sign Language (ASL) in Seattle and strictly connected to the Pro-Tactile Deaf-Blind empowerment movement (Edwards 2017, Granda and Nuccio 2018), Pro-Tactile is now expanding beyond the US, for instance in France, where a small but active community of Pro-Tactile French Sign Language (LSF) is growing. In their Pro-Tactile manifesto, Granda and Nuccio (2018) indicate seven general principles that represents the signature of Pro-Tactile. Some are cultural related, like principles 5, 6 and 7 and concern how to manage conflicts among other principles, always including the source of the information, and indicating sharing experience as more relevant than just transmitting information. Other principles reflect linguistic aspects like the use of space, which cannot correspond to the empty area in front of the signer, as in visual and tactile sign languages, but rather it must involve some solid surface (e.g., the upper part of the thigh, or the upper part of the chest of the addressee); the principle governing the modality used during an exchange, requiring both signers in a conversation to communicate tactually, while with tactile sign languages if one interlocutor is sighted the Deaf-Blind signer would use the visual sign language; finally the principles governing the use of classifiers, which again prescribe contact with the body of the addressee, hence tactile feeling, rather than articulation in the empty space between two signers. Thus, one of the distinctive features of pro-tactile languages is that their articulatory system exploits for communicative purposes the fact that the hands of the addressee touch those of the signer and that the body of the addressee is in the proximity. This exploitation is visible in Figure 2 from Pro-Tactile LSF, where the addressee distal part of the arm and hand stand for the wolf character in Lille Red Riding Hood story (red circle) and the hand of the signer represents the hunter walking towards him (cyan circle).

Figure 2: Pro-Tactile LSF. The signer on the right uses the body of the addressee to co-articulate part of a narration.



In addition to hand and arm of the addressee, also locations on the addressee's body can be employed as part of the articulatory options. So, for instance, the upper part of the thigh or the part above the chest of the addressee can be used as the location where classifiers can be positioned.²

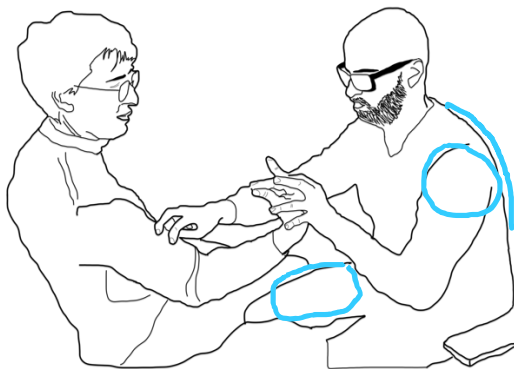
Interestingly, similar constructions have been independently documented in Tactile Norwegian and Swedish Sign Language (Mesch et al. 2015), indicating perhaps autonomous evolution due to external pressures introduced by blindness, independent from the Pro-Tactile social empowerment movement (see below Figure 8). Zorzi et al. (2025) document a similar use of space in the interpreting practice especially in the context of role-shift for Tactile Norwegian Sign Language. The authors use the term “tactile sign language haptics” to refer to this use of the signing space.³ The authors argue that these are signs whose point of articulation shifts from the neutral signing space or the signer's body (in this case, the interpreter) to the body of the Deaf-Blind addressee, and contrast with more “social haptic communication haptics”. In a broader sense, haptic coding is a conventional tactile-based system of gestures used to convey, among other things, emotions and environmental information about people present in the vicinity, coming, leaving, approaching, etc (i.a., Lahtinen and Palmer 2012, Volpato 2023, 2024). Zorzi and Volpato (2025) illustrate how fluent Italian tactile signers can integrate haptic gestures in the tactile sign stream. This integration can happen by borrowing haptic signs as if

² This is possible only if both signer and addressee agree that that body-part can be used.

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they were actual signs of the tactile sign language, but also by incorporating only some sub-components. For example, the haptic system uses body parts that are not normally used in tactile sign language, like the upper part of the thigh of the addressee, the shoulder, or the back (see figures 3 and 5). Haptic gestures articulated on the back can be produced by a third person, whose conversational role is specifically that of informing the Deaf-Blind person about the environment. The back is also used to provide information while the Deaf-Blind person is moving.

Figure 3. Favorite locations for haptic gestures: upper part of the thigh, shoulder, and back.



Interestingly, Italian tactile signers can use “haptic locations” to articulate classifier signs that would be normally produced in the empty space in front of the signer in visual LIS and in tactile LIS (Grigi 2025). Zorzi and Volpato (2025) argued that this use of the body somehow frees articulatory options that would otherwise be unavailable were tactile LIS to maintain the phonological inventory of visual LIS.

Given these premises, one may speculate that Scandinavian tactile signers, who regularly use the haptic system since long time, have reached a point in which parts of the body of the addressee are incorporated into the phonological space of their tactile sign language, hence independently instantiating phenomena of joint articulation where both signer and addressee contribute to some sub-components of a sign, in a similar way as Pro-Tactile ASL signers do. This option would be only available to fluent signers of tactile LIS if they are also regular users of the haptic system.

3. Visual-gestural vs. tactile-gestural: more than just a modality transfer

While defining the main properties of visual, tactile, and pro-tactile languages, we briefly touched on some innovations which led to the differentiations amongst these language types. We would like to address here, comparatively and in a more systematic way, three major aspects where the modality shift is producing the most observable changes: improving communication effectiveness, preserving/neutralizing grammatical contrasts, and widening meaning transmission. In one way or another, these innovations exploit tactile resources subserving them to both broad and narrow linguistic purposes.

In the transition from the visual to the tactile modality, the linguistic system is affected at all levels. Several characteristics of how language is structured and used become inaccessible to Deaf-Blind people. The most compelling one is access to the non-manual components of visual sign languages as first noticed in Collins and Petronio (1998). In such a scenario, a number of possibilities may realize: the minimal possibility is that nothing is done to recover the specific feature or property, hence we observe the loss of that property (language impoverishment); another possibility is that tactile users become aware of the potential loss and introduce an innovation that allow them to reproduce in the tactile modality the function of the visual feature (adaptation strategy); a third possibility realizes when tactile users autonomously develop a new tool either to encode a new property or to widen an existing one (innovation strategy). We are going to see how adaptations and innovations have been implemented at various levels in tactile languages, starting with those improving communication effectiveness, then considering those involving preservation/loss of grammatical contrast, and finally those used to widen the expressive power of the tactile modality.

Improving communication effectiveness. One macroscopically observable adaptation concerns the very mode of communication. As discussed above, tactile communication may happen in two ways, the monologue mode, or the dialogue mode (see figure 1 above). The way in which the hands and the bodies are positioned have been extensively described in the literature (see i.a., Collins and Petronio 1998, Mesch 2001, Schwartz 2009, Volpato 2023, 2024) and responds to the most essential requirement of communication, namely the need for the message to be effectively delivered. However, one neglected consequence of this adaptation in its impact on one communication property of human language, namely broadcast transmission (Hockett 1960). In fact, both spoken and visual sign languages allow one-to-many transmission: the delivery can be performed by individuals in front of a large audience. Furthermore, spoken and

visual sign languages do not even require physical proximity, especially after visual communication has become a reality via mobile devices. On the other hand, the tactile modality imposes physical proximity between the two interlocutors and it is realized in a one-to-one manner.⁴ Indeed, sighted people, can partially attend a tactile conversation but they cannot take part to it, unless breaking the conversation and replacing one of the two interlocutors. One notable exception is represented by pro-tactile. In this system, it is possible to use a variant of the monologue position to let a signer simultaneously communicate with two signers, as illustrated in figure 4.

Figure 4. One-to-two communication in Pro-Tactile.



In this configuration, the signer (middle of the picture) uses both hands as dominant hands simultaneously, one for each addressee. In the case of asymmetric two-handed signs, namely signs that require both hands to be articulated and in which the dominant hand does not copy the non-dominant hand, the sign is repeated, once on the hand on one addressee and a second time on the hand of the other addressee (see [this video](#) on Pro-Tactile Theatre).

The message is transmitted simultaneously to two tactile interlocutors. Crucially, in doing this, the pro-tactile signers introduce another innovation, namely they break the asymmetry, typical of visual and tactile sign languages. In fact, visual sign languages are articulated with the dominant hand, typically using the non-dominant one either as a copy of the dominant hand, or as an articulatory support (Brentari 1998). This is true even when a visual signer shifts

⁴ An attempt to create a signing robot that could serve as a replicant puppet to allow long distant communication in tactile sign language has been engineered with the project PARLOMA (Russo et al. 2015). The project created a laboratory prototype of hand, arm and torso. Preliminary testing has been conducted but at the moment of writing the market version is unfortunately still far from becoming a reality.

dominance (e.g., when a right-handed signer uses their left hand as dominant during a stretch of discourse). Dominance can shift, but still one hand would be asymmetrically dominant, while the other would produce a copy of the dominant or serve as articulatory support. This configuration is even more constrained in tactile sign languages. Given the majority of right-handed people, tactile sign language is produced with the right hand as dominant and with the left hand as non-dominant (see Mesch 2001 for a larger treatment of handedness in tactile sign language). Left-handed signers typically adapt to this situation and use their right hand in production and their left-hand in perception. The simultaneous one-to-two communication enabled in Pro-Tactile introduces a major and autochthonous innovation in the language.

An interesting case is observed between the two Pro-Tactile LSF signers in figure 5. A visit to a scientific laboratory was filmed for an episode of the Deaf-TV show *l'œil et la main*. While communicating between themselves, the two Deaf-Blind signers were assisted by two Deaf interpreters who, provided social haptic information.

Figure 5. Pro-Tactile LSF and social haptic communication.



That the two signers are using Pro-Tactile LSF can be deduced by the fact that the signer (on the right) is using the upper part of the body of the addressee (on the left) as location to articulate part of the discourse, in according to the principles of pro-tactile (Granda and Nuccio 2018, and Section 2). The addressee has his right hand on the thigh of the signer to provide feedback. One interpreter is using social haptics to provide information on the back of the addressee, the other interpreter uses the deltoid of the signer to provide additional environmental information to the signer. Figure 5 shows compatibility between the use of a

pro-tactile language and social haptic gestures, leaving open the possibilities of Deaf-Blind signers to integrate them, if they so desire.

Turn taking is a domain where innovations have been introduced to foster communication in tactile languages. Several studies in a number of tactile sign languages now describe how conversation partners alternate and how feedback is given following the general principles of conversation analysis (i.a., Mesch 2000, 2001, Berge and Raanes 2013, Raanes and Berge 2017, Iwasaki et al. 2019). Interesting adaptations have been described in Mesch (2001) for Tactile Sweden Sign Language (STS, Svenskt teckenspråk), where alternations can be signaled by slightly moving the hands on the horizontal axis, while movement upwards signal the intention of keeping the conversation turn (also documented for Norwegian sign language Raanes 2011).⁵ Iwasaki et al. (2019) describe the interaction between haptic gestures and tactile Australian Sign Language (Auslan). Before the actual conversation starts, tactile signers negotiate a number of conversational parameters via the use of haptic gestures which replace most of phatic uses of language, including among the others, attention getter, feedback on the position of the interlocutor, and orientation of the bodies. In this respect, the haptic code, which has a language external origin, is recruited to convey information that in a visual sign language would be transmitted by eye-contact, head movements, and eye-gaze.

The interaction between the production and the perceptual systems is another domain where tactile innovations introduced a major shift with completely unexplored but potentially high impact on the architectural organization of language. Spoken and signed languages use completely independent systems for production and perception. Speech is produced manipulating the airflow in the vocal tract, while what is perceived with the ears are the acoustic waves. Similarly for visual sign language, the two systems are completely independent, the gestural system produces a visual signal which is processed by the eyes. Such separation is much less sharp in tactile sign languages. This is particularly transparent when the body of the addressee becomes part of the sign and it interacts with production system. In the Little Red Riding Hood example in figure 2 repeated here as figure 6, the distal part of the

⁵ Notice that manipulation of location along the vertical axis in the neutral space is used to convey specificity and/or domain restriction in several visual sign languages (i.a., Barberà 2015, Davidson and Gagne 2015). Were this grammatical feature to be maintained in these tactile languages, it might conflict with this aspect of turn taking, unless additional prosodic cues co-occur to distinguish the two functions. More generally, what happens when articulatory features are recruited in a tactile language with a different function than that they have in the visual correspondent is an unexplored territory see example (1) below for a lexical case in Tactile Auslan from Willoughby et al. (2020).

arm and the hand of the addressee correspond to the classifier for animal, the wolf character, while the proximal part of the arm is the path that the hunter, represented with a two-leg classifier by the signer, walks going towards grandma house (this is an example of what Edwards and Brentari 2020, 2021 define as proprioceptive objects).

Figure 6. The hand of the addressee is used to co-articulate part of the depictive scene.



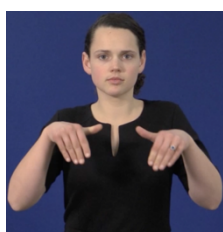
The interconnection with the body of the addressee serving at the same time as articulator and receptor has never been investigated from this perspective but opens interesting questions concerning the general organization of language, the division of labor between sentence building and discourse planning on the one hand, and their processing correspondents on the other hand. Not less important is the impact at the cognitive level, where attention, just to mention one aspect, is split between tasks concerning the effort to reach articulatory accuracy, the effort of perceiving the hand of the signer moving along the body, and the effort of understanding the global message, all happening at the same time. Further consequences of this aspect will be discussed in section 5, where the property of complete feedback is considered.

Preserving/loosing grammatical contrast. The linguistic properties of tactile sign languages attracted considerable attention in the past decades. In addition to language description, the interest focused on what happens to those properties of visual sign languages that are not accessible in the tactile modality. Specifically, since contrast is one of the key features around which human language grammar organizes its structure at all levels (phonology, morpho-syntax, and semantics), lack of perceptual access to certain features might induce considerable loss in contrast.

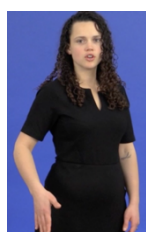
Although the number of studies is still small, interesting phenomena have been documented in tactile ASL (Collins and Petronio 1998), Aulsan (Iwasaki 2019, Willoughby et al. 2019, 2020),

LIS (Checchetto et al. 2018, Volpato 2023, 2024), LSF (Schwartz 2009), STS (Gabarró-López and Mesch 2020, Mesch 2001, 2011, 2013, Mesch et al. 2015), NTS (Mesch et al. 2015, Zorzi et al. 2025), and Pro-Tactile ASL (Edwards 2014a,b, 2015, 2017, 2018, Edwards and Brentari 2020, 2021). What emerges from this body of research is that different languages react in different ways to information loss. We discuss here potential cases of loss of contrast at the lexical level and at the sentential level.

An adaptation strategy used at the lexical level is documented for tactile Auslan (Willoughby et al. 2020), where data from a focus group indicate that signers seem to be aware of the possibility that some phonological features of place of articulation of visual Auslan might conflict with uses of the same feature for other purposes. Specifically, the authors document the case of DOG, which has three geographical variants in Auslan, as shown in (1) from the Auslan Signbank (<https://auslan.org.au>).



(1) a. DOG (variant 1)



b. DOG (variant 2)



c. DOG (variant 3)

The variant in (1a) is reported as used in the whole Country and is similar to the sign DOG in British sign language.⁶ The variant in (1b), also reported as used in the whole Country, is articulated on the upper part of the thigh. The variant in (1c) is produced in neutral space and is used in the north-east part of the Country. The variant in (1b) conflicts with the use of the upper part of the thigh for haptic purposes. The reflections from the focus group indicate that tactile Auslan signers prefer to adopt the northern variant in order to avoid the overlap of function of this location feature serving both as a location for lexical signs and a place for haptic signs.⁷ More generally, when multiple variants are available, Deaf-Blind signers prefer to use the one in the neutral space (Willoughby et al. 2020: 246). This solution clearly affects

⁶ The only difference is that all fingers including the thumb are selected in the Auslan variant, while the British one only has index and middle as selected fingers.

⁷ In the discussion, alternative strategies have also been proposed: one signer suggested to use the British sign language version of DOG, another to fingerspell the word *dog*.

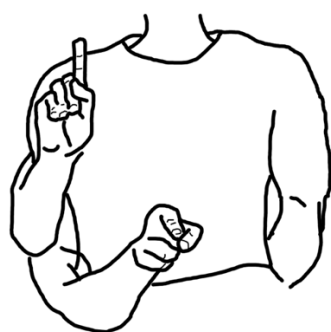
the distribution of contrastive features for place of articulation, and if it is systematically employed it may produce a reduction of the inventory of contrastive features.

A case in which contrast is preserved in an innovative way is represented by polar questions. Polar questions are often marked via eyebrow raising spreading on the whole sentence in visual sign language. A minimal pair with a simple declarative in visual LIS is given in (2), where the line above the glosses in (2b) indicates the distribution of the non-manual component (i.e., raised eyebrows).

- (2) a. MOM MOVIES GO
 ‘Mom went to the movies.’
 _____^{re}
 b. MOM MOVIES GO
 ‘Did mom go to the movies?’

The contrast in (2) is completely lost once switching to the tactile modality. This issue has been investigated for tactile ASL, LIS, and STS (Collins & Petronio 1998, Checchetto et al. 2018, Mesch 2001). Among the adaptation strategies, American tactile signers use a tag element, transform a direct question into a subordinate structure, or add the sign for “question”, as shown in figure 7.

Figure 7. The sign for question in ASL, used as a clause final particle to mark questions in Tactile ASL.



Italian tactile signers introduced an interesting linguistic innovation consisting in the use of the sign for *what* (with no mouthing) as a generic question-marker, extending its use to all questions, including yes/no questions. This strategy is illustrated in (3). In this example, the

sign WHAT—used to mark a content question in visual LIS—is used to mark a polar question in Tactile LIS.

(3) MOM SIGN WHAT SISTER SIGN **WHAT**

‘Did your mother or sister sign with you?’ (glossed [video](#))

This innovation is of high theoretical importance, since exactly the same mechanism of grammaticalization has been documented for spoken languages as well (König and Siemund 2007). The finding suggests that tactile sign languages are not simply transpositions into the tactile modality; rather, they may become autonomous, following a direction of change compatible with the options available to natural languages.

Furthermore, Checchetto et al. (2018) report that tactile interpreters were aware of the ambiguity reported in (2) and that borrowed the ASL strategy of using a “question mark” gesture at the end of a question. Crucially, Checchetto et al. (2018) collected data from pairs of Deaf-Blind signers and never found this strategy, neither in the corpus data or in the specific question-answer elicitation task.⁸

Notice that this simple move introduces one radical change in the way in which information is structured in the language. As noted in Checchetto et al. (2018), information is simultaneously layered in the visual sign language version of the sign, while it is sequentially disposed in its tactile correspondent. According to Checchetto et al. (2018), this phenomenon is more pervasive than it seems *prima facie*, affecting both the syntactic and the morphological levels.

Widening expressive power. The role of iconicity in human language has been at the center of a vivid debate (see Perniss et al. 2010 for a general overview), because it seems to be at odds with one of the foundational tenants of human language, namely its arbitrariness (i.a., De Saussure 1916, Hocket 1960, Chomsky 2009). On the other hand, visual sign languages are steeped in iconicity based on visual patterns and their mental representations (Cuxac 2000, Taub 2001, Liddell 2003). Iconic representations are observed in the lexicon (Perlman et al. 2019), in the morpho-syntax of classifier predicates (Benedicto & Brentari 2004, Grose et al.

⁸ Another strategy documented in Checchetto et al. was the use of a main clause interrogative verb. So direct questions were turned in to indirect questions. The two strategies, embedding the question in an interrogative predicate and the use of the sign WHAT, can co-occur.

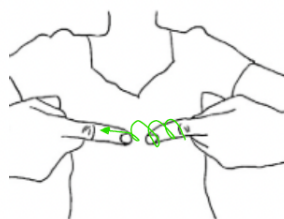
2007) and in the semantic structure of visual sign language (Schlenker 2022). The role of iconicity in spoken languages has been reconsidered from the functionalist perspective (i.a., Haiman 1985, Perniss et al. 2010), and the integration of gesture and speech as co-articulators of human language (Goldin-Meadow and Brentari 2017) is now paving the way to fully address the role of iconicity, especially in the semantic domain (for a detailed introduction to the semantics of iconicity see Schlenker 2020, 2022).

If iconicity is to be understood as an essential component of human language, parallel and integrated to its formal system, one should wonder what happens in tactile sign languages, where access to visual iconicity is completely lost. In explaining the “solid nature” of spatial representations for Deaf-Blind people, Edwards (2014a, 2017) reports the sign for PHONE CHARGER (figure 8a) in Pro-Tactile ASL. We discuss here only the part relative to the spiral cord. Visual iconicity would render the spiral cord as in figure 8b. The spiral shape of the cord is iconically mapped in visual sign language in a way that is not immediately perceivable by Deaf-Blind signers. In fact, it is represented by rolling the finger in the empty space. Pro-Tactile signers would place the same rolling movement inside the hand of the addressee, so that it can be tactilely perceived, as in (figure 8c).

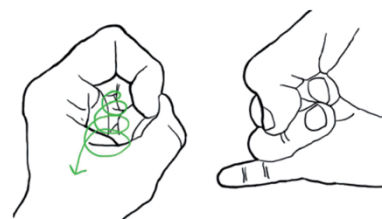
Figure 8a. Phone charger



8b. Visual iconicity



8c. Tactile iconicity



However, the depth of tactile iconicity is much wider and intersects the sentential and the discourse level, not just in pro-tactile languages, but more generally in tactile languages. One example is given in figure 9 from Mesch et al. (2015), where the Tactile STS signer uses the right hand of the addressee as a classifier representing a round object to identify a peach from a tree that has been previously introduced in discourse. The interesting aspect of these constructions is that the signer produces part of the content on the hand of the addressee. Specifically, in figure 9a she is describing the softness of the skin of a peach on a tree, as represented in the mental image in figure 9b. In other words, the hand of the addressee is not just moving and perceiving signs from the signer, it is part of the sign discourse itself.

Figure 9a. Addressee as signer



9b. Mental Representation.



Edwards (2017) documents one further aspect of how spatial reference also needs solid iconicity, as illustrated in figure 10. Figure 10a shows an interpreter (ineffectively) using a visual sign language pointing, figure 10b shows a tactile signer using the hand of the addressee as a spatial board pointing to a location that is the placeholder on a map previously introduced.

Figure 10a. Visual indexing



10b. Tactile indexing



Evidence reported in studies on Pro-Tactile ASL (Edwards & Brentari 2020) and Tactile STS and NTS (Mesch et al. 2015) indicates that visual schemes and vision-based mental representations do not immediately transfer into the tactile modality either at the lexical level (see figure 7) or at the morpho-syntactic level (see figures 8-10).

The emergence of tactile iconicity shows the need of Deaf-Blind signers to maintain a high degree of iconicity in their grammatical system. However, tactile iconicity is not a bare adaptation of visual iconicity, it requires a substantial cognitive re-assessment of what is relevant to be transmitted via iconicity.

As already mentioned, tactile iconicity is not an exclusive property of pro-tactile languages, as evidence for independent co-evolution exists in Tactile STS and NTS, where no Pro-Tactile movement exists. This fact is emblematic of the resilient nature tactile sign languages, but it is evidence that iconicity is not ancillary to human communication, it is rather deeply

systematized and interconnected with the formal components of human languages as also argued in Perniss et al. (2010). If iconicity were not so essential, there would be no reason for Deaf-Blind signers to re-code it in the tactile modality. Notice that this development of tactile iconicity is not immediate (i.e., it is not the result of transposition), but it is an emergent property of tactile languages. Understanding the conditions under which tactile iconicity operates may shed light on the connection between modality pressures and the computational system of human language.

One last property worth being discussed is role-shift, namely the possibility for signers to assume on their body the role of a participant described in the event (Schlenker 2017a, b).⁹ This is another angle where iconicity deploys its feature in sign language.

That role-shift as produced in visual sign languages may constitute a problem for tactile languages is a known issue (Granda and Nuccio 2018). In their programmatic work for Pro-Tactile ASL, the authors indicate that contact space (and classifiers) are used to univocally attribute the shifted role. For instance, the upper part of thigh of the addressee can be used as contact space where to locate the participants to the event described in the linguistic expression. A classifier or a pointing sign is used to establish the referential loci for these participants in the contact space (similar to what happens on the arm of the signer in figure 6 in Pro-Tactile LSF and on the hand of the signer in figure 9b in Pro-Tactile ASL), then whatever is expressed after a participant is retrieved in the contact space is to be understood as under role-shift. This is an innovative strategy not present in visual ASL. However, notice that the use of a manual indicator for role-shift is not unprecedented. In fact, signers of visual Japanese Sign Language can use a special manual sign to introduce role-shift (Kawasaki 2024).

To summarize, tactile and pro-tactile languages are not a bare adaptation of visual sign languages. They developed independent strategies to cope with potential information loss. Some of these strategies are adaptive, others are innovative. It is in these innovations (communication setting, encoding of phatic uses, turn taking negotiation, sequentialization of the message, grammaticalization of special signs, and tactile iconicity) that we observe the resilience nature human language witnessed in the tactile modality. Crucially, these innovations are manifestations of deeper architectural properties of human language.

⁹ The same phenomenon goes under the name of constructed dialogue and constructed action in the functionalist tradition (i.a., Liddell, 2003)

4. Tactile signers and tactile communities

In the previous sections, we defined the main properties of tactile and pro-tactile languages and we compared them with those of spoken and visual sign languages. In doing this, we mentioned that users of tactile languages are both deaf and blind, and that blindness is often an acquired and progressive condition on a pre-existing deafness condition. This is the typical situation of Deaf-Blind adults who have type 1 and 2 Usher syndrome, which is a genetic condition resulting in a congenitally profound or severe deaf person progressively losing sight (Otterstedde et al. 2001, D'esposito et al. 2025). The degree of vision loss varies from individual to individual causing functional blindness although it does not necessarily result in complete blindness. Also, the way in which vision is lost is not uniform. Usher people may lose sight by progressively restricting the visual field, hence the typical tunnel vision. However, Usher syndrome is not the only cause of deaf-blindness, indeed there are other etiologies for deaf-blindness both acquired (Alport syndrome, age-related sensory loss, injuries like strokes and brain damages, etc.) and congenital ones (prematurity, CHARGE syndrome, etc.). Depending on the etiology, deaf-blind people may become Deaf-Blind signers because this is the only language they can be exposed to (e.g., congenital cases), or because they mastered a visual sign language as first language and then they switch to the tactile modality once blindness does not allow the use of the visual sign language anymore. In this latter case, Deaf-Blind people typically switch from the visual sign language they acquired during their childhood to the tactile sign language in adulthood. It is possible that the various sources of deaf-blindness may interact with the process of acquiring and learning a tactile language. However, to our knowledge no study has addressed this specific aspect in any systematic way.

One interesting point, though, is the social correlation of identifying deaf-blindness with its etiology. Scoping reviews of the deafblind literature indicate that the medical view largely determines how Deaf-Blind people are identified (Simcock 2017, Jaiswal et al. 2018, Warnicke et al 2020). Classifications generally impose macro-categories like congenital deaf-blindness, congenital deafness followed by acquired blindness and vice-versa, or else specifically targets the particular conditions determining deaf-blindness (Dalby et al. 2009a, Otterstedde et al. 2001). Indeed, Deaf-Blind people are often grouped together with Deaf-Disabled even in sociological works (as reported in Leigh et al. 2020, 2023). In this respect, Deaf-Blind people are mostly seen as a category of people requiring a considerable amount of assistance (Dalby et al. 2009b), and this is precisely what major associations assisting deaf-blind people do. Unfortunately, this predominant view prevents addressing the real sociological condition of

Deaf-Blind people. One study that does not start from a medical view on deaf-blindness is Arndt & Parker (2016). They used semi-structured interviews and follow-up discussions with ten American deaf-blind people who volunteered to participate in the study during a one-week deaf-blind Retreat. The study pointed out several dimensions that allow a better identification of the social needs of Deaf-Blind people. These are: communication, mobility, work, and social interactions. In each of these aspects, Deaf-Blind participants experienced limitations, frustration, and a generally negative attitude, especially when dealing with non-deafblind people, as also emerged from the broader scoping literature review papers (Simcock 2017, Jaiswal et al. 2018, Warnicke et al 2020). On the other hand, the study also identified resilience and self-advocacy as fundamentally positive traits. This was particularly true for the users of Pro-Tactile ASL. Another aspect that emerged from this study is the benefit of collaborative peer-to-peer environments. Similar findings have been reported in the context of micro-social communities like families. Björk et al. (2022) conducted semi-structured interviews of non-deafblind partners in families in which the other partner is affected by deaf-blindness. Results indicate the presence of communication barriers in almost every situation, compensation for the deaf-blindness and the need of a great deal of support. In France, Schwartz (2009a) reports that the estimate of deaf-blindness is between 5600 and 6600 individuals for 2008 (National Centre for Deaf-blindness). This relatively large number includes all types of deaf-blind individuals that are covered by the medical definition. Schwartz also describes a situation in which deaf-blind people, including the users of Tactile LSF, do not seem to have developed a strong sense of community, evidenced by the fact that even the adaptation from visual to tactile sign language is carried out by sighted personnel and not by Deaf-Blind instructors. This situation seems to be in rapid change as now Pro-Tactile LSF is spreading both as a linguistic system and as an attempt to reach social empowerment within the Deaf-Blind community, as for instance promoted by the Deaf-Blind association [AccessVisuel](#). The situation of Australian Deaf-Blind people is reported to have similar dynamics to that documented in Seattle, although Deaf-Blind signers are not organized in a specific movement (Manns et al. 2022:4).

Although Deaf-Blind signers are found in many countries, most of the studies focus on Auslan, EU, and US tactile sign languages. One crucial difference between the Australian and European Deaf-Blind communities on one side and the American one on the other (or at least a considerable part of it) is the emergence of an empowerment movement led by Deaf-Blind people in the US. This is the Pro-Tactile movement (Edwards 2012, 2014a). Due to a number of circumstances, a large number of Deaf-Blind people created a solid network in the city of

Seattle. This allowed the consolidation of social empowerment leading to an overall better quality of life (Edwards 2014a), and the evolution of Tactile ASL towards innovative endogenous forms described in section 2 and 3. These forms do not have any correspondents either in sign or spoken language up to the point that mutual intelligibility between visual and tactile ASL has become difficult (Edwards and Brentari 2020). Notice that large European Deaf-Blind communities like the Norwegian and the Swedish ones do not seem to have initiated any empowerment movement similar to the American Pro-Tactile.

5. The status of languages in the tactile-gestural modality

Tactile and pro-tactile sign languages differ from the visual sign languages from which they originated in the way in which they are produced and perceived. This fact is leading to autonomous innovations, indicating that these systems are alive and evolve, in part as the result of pure modality adaptations, in part to fulfill grammatical purposes, and in part as a response to social needs. In the case of the Pro-Tactile American community, the innovations are making the languages diverging up to the point that mutual intelligibility between ASL and Pro-Tactile ASL has become very difficult (Edwards and Brentari 2020). It is then legitimate to reflect on what kind of systems tactile and pro-tactile sign languages are, and what their status is with respect to visual sign language and more generally as linguistic systems. More specifically, are languages in the tactile modality independent languages? are they dialectal variants of the visual sign language, or are they systems that use linguistic forms but not languages in the strict sense?

Depending from the specific angle one takes and the definition of what count as language, the answer could be different. As a pure Academic exercise, one could start to look for an answer by assessing Hockett's design features (Hockett 1960) and see whether tactile languages have them. Table 1 summarizes these properties, comparing visual, tactile, and pro-tactile languages.

Table 1. Hockett (1960) Design feature of human language.

<i>Design Feature</i>	<i>Visual SL</i>	<i>Tactile SL</i>	<i>Pro-Tactile SL</i>
<i>Vocal auditory channel</i>	No	No	No
<i>Broadcast transmission</i>	Yes	No	No
<i>Transitoriness</i>	Yes	Yes	Yes
<i>Interchangeability</i>	Yes	Yes	Yes

<i>Complete Feedback</i>	No	?	?
<i>Specialization</i>	Yes	Yes	Yes
<i>Arbitrariness</i>	Yes	Yes	Yes
<i>Discreteness</i>	Yes	Yes	Yes
<i>Displacement</i>	Yes	Yes	Yes
<i>Productivity</i>	Yes	Yes	Yes
<i>Traditional transmission</i>	Yes	Yes	Yes
<i>Duality of patterning</i>	Yes	Yes	Yes
<i>Prevarication</i>	Yes	Yes	Yes
<i>Reflexiveness</i>	Yes	Yes	Yes
<i>Learnability</i>	Yes	Yes	Yes

Besides channel, broadcast transmission, and complete feedback, a glance at table 1 shows that tactile languages have all the foundational properties of human language. A fortiori the same is true for visual sign languages, where only the channel and feedback cells are negative.

Of course, research in the past sixty years showed that human language is not limited to the vocal-acoustic modality, hence that property is not to be considered foundational anymore.¹⁰ If language can emerge in the visual-gestural modality, one can legitimately expect it to be conveyed also in the tactile-gestural modality. The rationale behind this expectation is that the core properties of human language are part of a cognitive system that can adapt and use whatever is suitable for externalization. The very existence of tactile languages, with the properties researchers are discovering, suggests that tactile perception adequately fulfills the cognitive demand for language use.

Complete feedback refers to the possibility of on-line self-perceiving, controlling, and modifying the production output, that is, the ability of a user to operate on the message while producing it. Visual sign languages can be controlled and modified on-line, but they cannot be completely self-perceived using the visual modality. This is because differently from speakers, who can hear what they produce vocally because of the acoustic properties of speech, signers

¹⁰ Of course, we ignore here written forms of language as well as tactile writing systems, as these are normed transpositions of spoken languages.

do not systematically look at what they are signing. Even if in principle signers could look at their hands while signing, they would still not have access to the non-manual components. Specifically, while speakers perceive all the prosodic features as expressed by all formants, signers do not have self-access to the prosodic information produced by their facial expressions while signing. However, if the purpose of self-perception is to monitor the production, there is psycholinguistic evidence that signers do that (Emmorey et al. 2009, Riès et al. 2020). Interestingly, Emmorey et al. (2009) suggest that the monitoring phase might involve somatosensory aspects.

The issue is completely different for tactile languages. The articulators touch the hands and the body of the addressee, hence tactile signers continuously have perceptual feedback of what they are producing, like a “negative impression” taken from the contact of their hands with the addressee’s. This is the reason of the question mark for tactile and pro-tactile languages cells in table 1. Tactile signers do have continuous and complete feedback because touch is reciprocal between addressee and signer, however, it is not exactly the same feedback as perceived by those who receive the message, a point that makes the quality of complete feedback different between the acoustic and the tactile modality. The situation is even more intricate for the cases of co-productions illustrated in figures 6 and 9 above, in which signer and addressee co-construct the sign. In these cases, each hand provides a qualitatively different kind of feedback to the signer. The dominant hand receives the negative impression feedback as described above, like in standard tactile production, however the non-dominant hand receives positive feedback as if it were in the listening mode. To illustrate, in figure 6 above, the dominant hand of the signer (on the right), while actively producing the sign for the hunter, is also receiving negative impression feedback because the tips of the fingers are touching the proximal part of the arm of the addressee. At the same time, the non-dominant hand is in the receiving mode and receives positive feedback from the distal part of the arm and hand of the addressee, who is co-signing the wolf.

Languages in the tactile modality do not have the broadcasting feature. This is a clear consequence of the one-to-one exchange setting, or eventually one-to-two in the case of Pro-Tactile. Certainly not enough to claim that Pro-Tactile has broadcasting transmission. However, as witnessed by the vast literature on animal communication, especially the one about monkey calls, broadcasting transmission is hardly an exclusive property of human language (i.e., Lemasson et al. 2010 for monkeys and Templeton et al. 2005 for birds).

At the end of this excursus, if we use Hockett's foundational features as a check list to detect human language, one might be tempted to say that both visual and tactile languages are natural languages. While the conclusion is indisputable because based on solid evidence build on decades of scientific research for visual sign languages, the same cannot be said for languages in the tactile modality, as one could in principle argue that these properties could depend on tactile languages originating from visual sign languages. In other words, tactile languages would inherit the properties they have because they are dialects of the corresponding visual sign language. Still, visual and sign languages differ in two of these features, namely broadcasting and complete feedback.

If this is the case, then the question is whether the derived tactile language has enough linguistic distance to be considered as an independent system. This is an empirical question that cannot be addressed globally, but only looking case-by-case to specific pairs of visual/tactile languages. Interestingly, we observe a similar direction in the differentiation from visual sign language in the use of the addressee's body to construct part of the discourse across several unrelated languages (Pro-Tactile ASL and LSF but also Tactile STS and NTS, and perhaps emerging in Tactile LIS). A similar direction is also observed in the exploitation of tactile iconicity, perhaps more prominently in pro-tactile languages. We also observe cases in which the differentiation is taking opposite paths. For instance, pro-tactile languages refrain from using the neutral space as the locus of signs, especially of classifier constructions (see the notion of contact space provided in Granda and Nuccio 2018), while Tactile Auslan, for instance, uses signs in that area for variants that would otherwise conflict with places of articulation used for negotiating other aspects of the communication (Willoughby et al. 2020). It is intriguing that the same linguistic request of clarity is answered in two completely opposite ways and in a sense, it is also suggestive of true and independent linguistic systems. Indeed, the grammar of spoken languages often reacts in opposite ways to the very same constraint. For instance, languages may use epenthesis or deletion to avoiding offending consonant or vowel clusters. Turning back to the issue of linguistic distance between visual and tactile languages, we should also notice that the time span separating the two is generally very small, probably a matter of few decades (Willoughby et al. 2018). While this is hardly enough to let two spoken languages diverge up to the point that they can be considered two different languages, the rate of linguistic change seems to be quite faster for visual signed languages (Abner et al. 2024), hence leaving the verdict open for tactile languages. In addition to this,

one should consider that tactile sign languages are relatively isolated from visual sign languages in the sense that only Deaf-Blind people with enough visual residue can productively use both systems. Systematic and direct contact between the two modalities among Deaf-Blind people is thus quite rare. It is true that Deaf-Blind people interact a lot with sighted signers who regularly use the visual sign language (e.g., Deaf friends, care givers, interpreters, etc.), however, contact with the visual sign language is never direct in these cases.¹¹ This is even more extreme in the case of Pro-Tactile ASL, because when engaging in a conversation with a sighted signer, the entire conversation must remain tactile by explicit choice (Granda and Nuccio 2018). Specifically, a sighted signer would receive the message in Pro-Tactile ASL and not in visual ASL. As already mentioned above, preliminary evidence suggests that Pro-Tactile ASL is becoming further and further independent from visual ASL up to the point that mutual intelligibility is heavily affected. Indeed, lacking a comprehensive comparative assessment of the grammar of visual and tactile sign languages, mutual intelligibility could be used as a screening criterion to evaluate the distance between a visual and a signed language. In this case, one might consider Pro-Tactile ASL independent enough to be treated as an independent language from (visual) ASL.¹²

On the other hand, using this criterion, if two languages are mutually intelligible, then they should not be considered as independent languages. The question then is what kind of variant/dialect are tactile languages in this situation. One interesting angle to explore is whether they would qualify as different registers of the same language. If the medium were the only factor determining the difference, this would be an attractive possibility. However, the sociology of Deaf and Deaf-Blind people seems to indicate that considering the two systems as register variants would be too much reductive (see the discussion in Section 4). Interestingly, this seems to be the attitude expressed by some members of the Italian Deaf community. Indeed, the Italian law recognizes at the same time LIS and Tactile LIS as the languages of the Deaf and Deaf-Blind communities. While this was happening in the Italian parliament, in a public Vlog discussion held in May, 2021¹³, Tactile LIS was considered the same as (visual)

¹¹ How language contact might work in this scenario is a completely unexplored territory. One can imagine that the biases from the visual sign language might affect the tactile sign language used by the sighted signer hence putting pressure on the tactile sign language used by the Deaf-Blind interlocutor. While there are studies on the interaction between interpreters and Deaf-Blind users, to our knowledge there are no studies specifically addressing the contact issue.

¹² This is also corroborated by several points in which the grammar of Pro-Tactile ASL differs from that of ASL, including the nature of morpho-phonological units that are targeted by grammatical processes (Edwards and Brentari 2020).

¹³ POLIS – Dossier LIS storia di un percorso lungo 40 anni (Polis – LIS dossier: the story of a 40-year journey)

LIS and as a tool to support Deaf-Blind people by one of the Deaf discussants. So at least among Deaf signers the knowledge exists that Tactile LIS is similar enough to visual LIS, but that it is also different as Tactile LIS serves other (more utilitarian) purposes. Whether the claim that tactile sign languages should be treated as a mere utilitarian variant of visual sign languages is true is quite unlikely, as evidence from Tactile LIS itself and other tactile languages indicate that these systems do develop independent grammatical features and are more than just tools (see Section 2 and 3). Furthermore, a similar utilitarian approach is observed when a visual sign language is used in educational settings, like bilingual bimodal programs, with the purpose of giving sign language input to deaf children that do not have signing parent. In other words, whether a language is used in a utilitarian way or not, its status as a language does not change.

One final point concerns language transmission. While tactile languages can be learned, they are not normally transmitted by simple exposure as first language. Specifically, native users are virtually absent. The incidence of congenital deaf-blindness (1 in 29.000, according to the [National Center on Deafblindness](#)) is much smaller compared to the incidence of congenital deafness (about 1.5 in 1000, Yoshimura et al. 2024); even if the precise statistics about the number of deaf-blind children born in families regularly using a tactile language is unknown, the proportion is by far smaller than the well-known 5% of deaf children born in signing families (Mitchell and Karchmer 2004). In this respect, tactile languages are almost exclusively a second language, something that is acquired after a visual sign language is already fully acquired. Is this enough to exclude tactile languages from being natural languages? If lack of native users is a necessary condition for a language to be considered as such, then tactile sign languages do not characterize as typical human languages. However, evidence that this might be a too strict conclusions come from the study of grammatical competence of non-native signers. Zorzi et al. (2022) shows that non-native signers of visual LIS have developed a robust and consistent, but slightly different grammar from native signers.¹⁴ Furthermore, we should also notice that the situation of Deaf-Blind communities is in constant development as tactile languages are becoming more and more popular as an effective communication system for Deaf-Blind people. The scientific interest in their features also contributes to increase their perceived status and they are more and more used as an intervention tool for congenitally born

¹⁴ In some respect, the situation might not be different from the grammatical competence of creole speakers (Aboh 2015, 2020).

deaf-blind. So, the number of congenital deaf-blind children that are exposed to a tactile language as their first language will increase constantly.

6. Conclusions and remarks for future work

Tacking stock, either directly inherited, or adapted, or innovated from visual sign languages, tactile languages seem to have all the key features that human languages have, combinatorial properties included. When research on specific properties is available, the results indicate that tactile languages are differentiating from visual sign languages, on their way to become autonomous languages. If they are more and more used in intervention situations especially with congenital deaf-blind children, they would also have a larger and larger number of L1 signers. In parallel to the linguistic development, Deaf-Blind signers are becoming aware of the potentials of their language and they are proud to use and advocate for them, as witnessed by the Pro-Tactile movement in the US.

While we cannot provide a generalized conclusion that all tactile and pro-tactile languages should be granted the status of independent languages from their visual counterparts, there seems to be ample evidence to treat them as an independent human language type.

Future research should study the properties of tactile languages more systematically, so that we can better assess the dimensions in which the different modalities may unveil core properties of human language. Cross-modal comparative studies have already established that it is not correct to equate sign and spoken languages, but rather that sign languages should be equated to spoken languages plus the gestural system as far as expressive power is concerned (Goldin-Meadow and Brentari 2017). Our study highlights that it is too limited to understand iconicity as an ancillary property of human language, mostly exploited by the visual modality; rather, iconicity seems to be a more basic property of how human cognition chooses to transmit meaning when possible.

The more we know about tactile languages, the more we can inform society and policy makers on how to address the communication issues of Deaf-Blind people. The interaction between stakeholders (e.g., Deaf-Blind associations) and Academic centers is essential in this goal. For instance, the joint contribution of Lega del Filo d'Oro Foundation (the most important Italian association of Deaf-Blind people) and linguist group from the University of Milano-Bicocca and Ca' Foscari of Venice allowed to recognize Tactile LIS at the same level as visual LIS.

Finally, it is particularly important that a positive attitude is built around tactile and pro-tactile languages and most importantly that wrong opinions (e.g., tactile languages are just tools for deaf-blind people) are amended with science-based evidence.

References

- Abner, Natasha R., Grégoire Clarté, Carlo Geraci, Robin J. Ryder, Justine Mertz, Anah Salgat & Shi Yu (2024): Computational phylogenetics reveal histories of sign languages. *Science* 383(February). 519–523. doi:[10.4324/9780367854850-21](https://doi.org/10.4324/9780367854850-21).
- Aboh, Enoch O. (2020): Lessons From Neuro-(a)-Typical Brains: Universal Multilingualism, Code-Mixing, Recombination, and Executive Functions. *Frontiers in Psychology* 11. 1–16. doi:[10.3389/fpsyg.2020.00488](https://doi.org/10.3389/fpsyg.2020.00488).
- Aboh, Enoch Oladé (2015): *The Emergence of Hybrid Grammars: Language Contact and Change*. 1. edn. Cambridge University Press. doi:[10.1017/CBO9781139024167](https://doi.org/10.1017/CBO9781139024167).
- Arndt, Katrina & Amy Parker (2016): Perceptions of Social Networks by Adults Who Are Deafblind. *American Annals of the Deaf* 161(3). 394–403.
- Baker, Anne, Beppie Van den Bogaerde, Roland Pfau & Trude Schermer (2016): *The linguistics of sign languages*. John Benjamins Publishing Company.
- Baker, Anne & Bencie Woll (2008): *Sign Language Acquisition*. (Ed.) Ann Baker & Bencie Woll Amsterdam: John Benjamins Publishing Company.
- Barberà Altimira, Gemma (2015): *The Meaning of Space in Sign Language: Reference, Specificity and Structure in Catalan Sign Language Discourse*. DE GRUYTER. doi:[10.1515/9781614518815](https://doi.org/10.1515/9781614518815).
- Benedicto, Elena & Diane Brentari (2004): Where did all the arguments go? Argument-changing properties of classifiers in ASL. *Natural Language & Linguistic Theory* 22. 743–810.
- Björk, Maria, Moa Wahlqvist, Karina Huus & Agneta Anderzén-Carlsson (2022): The consequences of deafblindness rules the family: Parents' lived experiences of family life when the other parent has deafblindness. *British Journal of Visual Impairment* 40(1). 18–28. doi:[10.1177/0264619620941895](https://doi.org/10.1177/0264619620941895).
- Cecchetto, Alessandra, Carlo Geraci, Carlo Cecchetto & Sandro Zucchi (2018): The language instinct in extreme circumstances: The transition to tactile Italian Sign Language (LIST) by Deafblind signers. *Glossa* 3(1)(66). 1–28. doi:[10.5334/gjgl.357](https://doi.org/10.5334/gjgl.357).
- Chomsky, Noam (2009): *Cartesian Linguistics*. Cambridge, UK: Cambridge University Press.

- Collins, Steven Douglas & Karen Petronio (1998): What happens in Tactile ASL? In Ceil Lucas (Hrsg.), *Pinky Extension and Eye Gaze: Language Use in Deaf Communities*, 18–37. Washington, DC: Gallaudet University Press.
- Cuxac, Christian (2000): *La langue des Signes Française (LSF) Les voies de l'iconicité*. Ophrys.
- Dalby, Dawn M., John P. Hirdes, Paul Stolee, J. Graham Strong, Jeff Poss, Erin Y. Tjam, Lindsay Bowman & Melody Ashworth (2009a): Characteristics of individuals with congenital and acquired Deaf-Blindness. *Journal of Visual Impairment and Blindness* 103(2). 93–102. doi:[10.1177/0145482x0910300208](https://doi.org/10.1177/0145482x0910300208).
- Dalby, Dawn M., John P. Hirdes, Paul Stolee, J. Graham Strong, Jeff Poss, Erin Y. Tjam, Lindsay Bowman & Melody Ashworth (2009b): Development and psychometric properties of a standardized assessment for adults who are deaf-blind. *Journal of Visual Impairment and Blindness* 103(1). 7–16. doi:[10.1177/0145482x0910300103](https://doi.org/10.1177/0145482x0910300103).
- Davidson, Kathryn (2020): Is experimental a gradable predicate? In Mariam Asatryan, Yixiao Song & Ayana Whitmal (Hrsg.), *NELS 50: Proceedings of the Fiftieth Annual Meeting of the North East Linguistic Society*, 125–144. Amherst, MA: GLSA.
- Davidson, Kathryn & Deanna Gagne (2022): “More is up” for domain restriction in ASL. *Semantics and Pragmatics* 15(1). 1–52. doi:[10.3765/sp.15.1](https://doi.org/10.3765/sp.15.1).
- De Saussure, Ferdinand (1916): *Course in general linguistics*. New York, NY: Philosophical Library.
- D’Esposito, Fabiana, Giuseppe Gagliano, Caterina Gagliano, Antonino Maniaci, Alessandro Avitabile, Rosa Giglio, Michele Reibaldi, Maria Francesca Cordeiro & Marco Zeppieri (2025): Usher Syndrome: New Insights into Classification, Genotype–Phenotype Correlation, and Management. *Genes* 16(3). 332. doi:[10.3390/genes16030332](https://doi.org/10.3390/genes16030332).
- Edwards, Terra (2014a): From compensation to integration: Effects of the pro-tactile movement on the sublexical structure of Tactile American Sign Language. *Journal of Pragmatics*. Elsevier B.V. 69. 22–41. doi:[10.1016/j.pragma.2014.05.005](https://doi.org/10.1016/j.pragma.2014.05.005).
- Edwards, Terra (2014b): *Language Emergence in the Seattle DeafBlind Community*. University of California, Berkeley doi:[10.1192/bjp.205.1.76a](https://doi.org/10.1192/bjp.205.1.76a).
- Edwards, Terra (2015): Bridging the gap between DeafBlind minds: Interactional and social foundations of intention attribution in the Seattle DeafBlind community. *Frontiers in Psychology* 6(OCT) doi:[10.3389/fpsyg.2015.01497](https://doi.org/10.3389/fpsyg.2015.01497).

- Edwards, Terra (2017): Sign-Creation in the Seattle DeafBlind Community: a triumphant story about the regeneration of obviousness. *Gesture* 16(2). 307–332. doi:[10.1075/gest.16.2.06edw](https://doi.org/10.1075/gest.16.2.06edw).
- Edwards, Terra (2018): Re-Channeling Language: The Mutual Restructuring of Language and Infrastructure among DeafBlind People at Gallaudet University. *Journal of Linguistic Anthropology* 28(3). 273–292. doi:[10.1111/jola.12199](https://doi.org/10.1111/jola.12199).
- Edwards, Terra & Diane Brentari (2020): Feeling phonology: The conventionalization of phonology in protactile communities in the United States. *Language* 96(4). 819–840. doi:[10.1353/lan.2020.0063](https://doi.org/10.1353/lan.2020.0063).
- Edwards, Terra & Diane Brentari (2021): The Grammatical Incorporation of Demonstratives in an Emerging Tactile Language. *Frontiers in Psychology* 11(January). 1–17. doi:[10.3389/fpsyg.2020.579992](https://doi.org/10.3389/fpsyg.2020.579992).
- Emmorey, Karen, Rain G. Bosworth & Tanya Kraljic (2009): Visual feedback and self-monitoring of sign language. *Journal of memory and language* 61(3). 398–411. doi:[10.1016/j.jml.2009.06.001](https://doi.org/10.1016/j.jml.2009.06.001).
- Emmorey, Karen & Melissa Herzig (2003): Categorical Versus Gradient Properties of Classifier Constructions in ASL. In Karen Emmorey (Hrsg.), *Perspectives on Classifier Constructions in Sign Languages*, 221–246. New York: Psychology Press.
- Gabarró-López, Sílvia & Johanna Mesch (2020): Conveying Environmental Information to Deafblind People: A Study of Tactile Sign Language Interpreting. *Frontiers in Education* 5. 1–12. doi:[10.3389/feduc.2020.00157](https://doi.org/10.3389/feduc.2020.00157).
- Gavrilescu, Robert, Carlo Geraci & Johanna Mesch (2024): Content Questions in Sign Language From theory to language description via corpus, experiments, and fieldwork. In Eleni Efthimiou, Stavroula-Evita Fotinea, Thomas Hanke, Julie A. Hochgesang, Johanna Mesch & Marc Schulder (Hrsg.), 86–94. Torino, Italy.
- Geraci, Carlo (2009): Real World and Copying Epenthesis: the case of classifier predicates in Italian Sign Language. In Anisa Schardl, Martin Walkow & Muhammad Abdurrahman (Hrsg.), *Proceedings of the North East Linguistic Society* 38, 237–250. Amherst: GLSA.
- Goldin-Meadow, Susan & Diane Brentari (2017): Gesture, sign and language: The coming of age of sign language and gesture studies. *Behavioral and Brain Sciences* 1–82. doi:[10.1017/S0140525X15001247](https://doi.org/10.1017/S0140525X15001247).
- Granda, Aj & Jelica Nuccio (2018): *Protactile Principles*. Seattle, Wa: Tactile Communication.

- Grigi, Giorgio (2025): Testimonianza sulla Comunicazione Socio Aptica (CSA). Gehalten auf der La comunicazione e la sordocità: LIS tattile e comunicazione socio-aptica, Milano, Italy.
- Grose, Donovan, Ronnie B. Wilbur & Katharina Schalber (2007): Events and telicity in classifier predicates: A reanalysis of body part classifier predicates in ASL. *Lingua* 117(7). 1258–1284. doi:[10.1016/j.lingua.2005.06.014](https://doi.org/10.1016/j.lingua.2005.06.014).
- Haiman, John (Hrsg.) (1985): *Iconicity in Syntax: Proceedings of a symposium on iconicity in syntax, Stanford, June 24–26, 1983* (Typological Studies in Language). Bd. 6. Amsterdam: John Benjamins Publishing Company. doi:[10.1075/tsl.6](https://doi.org/10.1075/tsl.6).
- Hockett, Charles F. (1960): The Origin of Speech. *Scientific American* 203. 88–97.
- Humphries, Tom, Poorna Kushalnagar, Gaurav Mathur, Donna Jo, Christian Rathmann & Scott Smith (2019): Support for parents of deaf children: Common questions and informed, evidence-based answers. *International Journal of Pediatric Otorhinolaryngology*. Elsevier. 118(November 2018). 134–142. doi:[10.1016/j.ijporl.2018.12.036](https://doi.org/10.1016/j.ijporl.2018.12.036).
- Humphries, Tom, Poorna Kushalnagar, Gaurav Mathur, Donna Jo Napoli & Carol A. Padden (2014): Ensuring language acquisition for deaf children: What linguists can do. *Language* 90(2). 31–52. doi:[10.1353/lan.2014.0036](https://doi.org/10.1353/lan.2014.0036).
- Iwasaki, Shimako, Meredith Bartlett, Howard Manns & Louisa Willoughby (2019): The challenges of multimodality and multi-sensoriality: Methodological issues in analyzing tactile signed interaction. *Journal of Pragmatics*. Elsevier Ltd. 143. 215–227. doi:[10.1016/j.pragma.2018.05.003](https://doi.org/10.1016/j.pragma.2018.05.003).
- Jaiswal, Atul, Heather Aldersey, Walter Wittich, Mansha Mirza & Marcia Finlayson (2018): Participation experiences of people with deafblindness or dual sensory loss: A scoping review of global deafblind literature. *PLoS ONE* 13(9) doi:[10.1371/journal.pone.0203772](https://doi.org/10.1371/journal.pone.0203772).
- Kawasaki, Noriko (2024): Syntax and semantics of Role Shift in Japanese Sign Language. *Sign Language & Linguistics* 27(2). 209–239. doi:[10.1075/sll.22008.kaw](https://doi.org/10.1075/sll.22008.kaw).
- Kimmelman, Vadim (2021): Acceptability Judgments in Sign Linguistics. In Grant Goodall (Hrsg.), *The Cambridge Handbook of Experimental Syntax*, 561–584. Cambridge, MA: Cambridge University Press. doi:[10.1017/9781108569620](https://doi.org/10.1017/9781108569620).
- Lahtinen, Riitta & Russ Palmer (2012): Environmental description. In Elisa Perego (Hrsg.), *Emerging topics in translation: Audio description*, 105–114. Trieste, IT: EUT.
- Leigh, Irene W., Jean F. Andrews, Cara A. Miller & Ju-Lee A. Wolsey (2023): *Deaf people and society*. New York: Routledge.

- Leigh, Irene W. & Catherine A. O'Brien (2020): *Deaf Identities*. (Ed.) Irene W. Leigh & Catherine A. O'Brien *Deaf Identities*. Oxford, UK: Oxford University Press. doi:[10.1093/oso/9780190887599.001.0001](https://doi.org/10.1093/oso/9780190887599.001.0001).
- Lemasson, Alban, Karim Ouattara, Hélène Bouchet & Klaus Zuberbühler (2010): Speed of call delivery is related to context and caller identity in Campbell's monkey males. *Naturwissenschaften* 97(11). 1023–1027. doi:[10.1007/s00114-010-0715-6](https://doi.org/10.1007/s00114-010-0715-6).
- Lettieri, Jessica, Mirko Santoro & Carlo Geraci (2023): On Elicited Data in Sign Language Syntax. *FEAST. Formal and Experimental Advances in Sign language Theory* 5. 88–99. doi:<https://doi.org/10.31009/FEAST.i5.08>.
- Liddell, Scott K. (2003): *Grammar, Gesture and Meaning in American Sign Language*. Cambridge, UK: Cambridge University Press.
- Mesch, Johanna (2001): *Tactile Sign Language. Turn-taking and questions in signed conversations of deaf-blind people*. Signum Verlag.
- Mesch, Johanna (2011): Variations in tactile signing - the case of one-handed signing. *Journal of Estonian and Finno-Ugric Linguistics* 2(1). 273–282.
- Mesch, Johanna (2013): Tactile Signing with One-Handed Perception. *Sign Language Studies* 13. 238–263. doi:[10.1353/sls.2013.0005](https://doi.org/10.1353/sls.2013.0005).
- Mesch, Johanna (2016): Sign Language: Tactile. In Genie Gertz & Patrick Boudreault (Hrsg.), *SAGE Deaf studies encyclopedia*, 820–822. Washington, DC: Gallaudet University Press.
- Mesch, Johanna, Eli Raanes & Lindsay Ferrara (2015): Co-forming real space blends in tactile signed language dialogues. *Cognitive Linguistics* 26(2). 261–287. doi:[10.1515/cog-2014-0066](https://doi.org/10.1515/cog-2014-0066).
- Mitchell, Ross E & Michael A Karchmer (2004): When Parents Are Deaf Versus Hard of Hearing: Patterns of Sign Use and School Placement of Deaf and Hard-of-Hearing Children. *Journal of Deaf Studies and Deaf Education* 9(2). 133–152. doi:[10.1093/deafed/enh017](https://doi.org/10.1093/deafed/enh017).
- Nyst, Victoria (2012a): Shared sign languages. In Roland Pfau, Markus Steinbach & Bencie Woll (Hrsg.), *Sign Language: An International Handbook*, 552–574. Mouton de Gruyter.
- Nyst, Victoria (2012b): Shared sign languages. In Roland Pfau, Markus Steinbach & Bencie Woll (Hrsg.), *Sign Language: An International Handbook*, 552–574. De Gruyter Mouton.
- Orfanidou, Eleni, Bencie Woll & Gary Morgan (2015): *Research Methods in Sign Language Studies*. Wiley-Blackwell.
- Otterstedde, Christian R., Ulrich Spandau, Anita Blankenagel, William J. Kimberling & Christoph Reisser (2001): A new clinical classification for Usher's syndrome based on a

- new subtype of Usher's syndrome type I. *Laryngoscope* 111(1). 84–86. doi:[10.1097/00005537-200101000-00014](https://doi.org/10.1097/00005537-200101000-00014).
- Padden, Carol A. & Tom Humphries (2005): *Inside deaf culture*. Harvard University Press.
- Perlman, Marcus, Hannah Little, Bill Thompson & Robin L. Thompson (2019): Iconicity in Signed and Spoken Vocabulary: A Comparison Between American Sign Language, British Sign Language, English, and Spanish. In Wendy Sandler, Marianne Gullberg & Carol A. Padden (Hrsg.), *Visual Language*, 132–147. Lausanne, CH: Frontiers Media. doi:[10.3389/fpsyg.2018.01433](https://doi.org/10.3389/fpsyg.2018.01433).
- Perniss, Pamela, Robin L. Thompson & Gabriella Vigliocco (2010): Iconicity as a General Property of Language: Evidence from Spoken and Signed Languages. *Frontiers in Psychology* 1 doi:[10.3389/fpsyg.2010.00227](https://doi.org/10.3389/fpsyg.2010.00227).
- Petitto, Laura Ann & Paula F. Marentette (1991): Babbling in the Manual Mode: Evidence for the Ontogeny of Language. *Science* 251. 1943–1946.
- Petronio, Karen & Valerie Dively (2006): Yes, #no, visibility, and variation in ASL and tactile ASL. *Sign Language Studies* 7(1). 57–98. doi:[10.1353/sls.2006.0032](https://doi.org/10.1353/sls.2006.0032).
- Quer, Josep, Carlo Cecchetto, Caterina Donati, Carlo Geraci, Meltem Kelepir, Roland Pfau & Markus Steinbach (Hrsg.) (2017): *SignGram Blueprint: A Guide to Sign Language Grammar Writing*. Berlin: De Gruyter Mouton.
- Reed, Charlotte M., Lorraine A. Delhorne, Nathaniel I. Durlach & Susan D. Fischer (1995): A Study of the Tactual Reception of Sign Language. *Journal of Speech, Language, and Hearing Research* 38(2). 477–489. doi:[10.1044/jshr.3802.477](https://doi.org/10.1044/jshr.3802.477).
- Riès, Stephanie K., Linda Nadalet, Soren Mickelsen, Megan Mott, Katherine J. Midgley, Phillip J. Holcomb & Karen Emmorey (2020): Pre-output Language Monitoring in Sign Production. *Journal of Cognitive Neuroscience* 32(6). 1079–1091. doi:[10.1162/jocn_a_01542](https://doi.org/10.1162/jocn_a_01542).
- Russo, Ludovico Orlando, Giuseppe Airò Farulla, Daniele Pianu, Alice Rita Salgarella, Marco Controzzi, Christian Cipriani, Calogero Maria Oddo, Carlo Geraci, Stefano Rosa & Marco Indaco (2015): PARLOMA – A Novel Human-Robot Interaction System for Deaf-Blind Remote Communication. *International Journal of Advanced Robotic Systems* 12(57). 1–13. doi:[10.5772/60416](https://doi.org/10.5772/60416).
- Sandler, Wendy (2012): The Phonological Organization of Sign Languages. *Linguistics and Language Compass* 6(3). 162–182. doi:[10.1002/lnc3.326](https://doi.org/10.1002/lnc3.326).
- Sandler, Wendy & Diane Lillo-Martin (2006): *Sign language and linguistic universals*. Cambridge, UK: Cambridge University Press.

- Santoro, Mirko (2018): *Compounds in sign languages: The case of Italian and French Sign Language*.
- Schlenker, Philippe (2017a): Super Monsters I: Attitude and Action Role Shift in Sign Language. *Semantics and Pragmatics* 1–44.
- Schlenker, Philippe (2017b): Super monsters II: Role Shift, Iconicity and Quotation in Sign Language. *Semantics and Pragmatics* 10(12). 1–67. doi:[10.3765/sp.10.9](https://doi.org/10.3765/sp.10.9).
- Schlenker, Philippe (2020): Gestural grammar. *Natural Language and Linguistic Theory*. Springer Nature B.V. 38(3). 887–936. doi:[10.1007/s11049-019-09460-z](https://doi.org/10.1007/s11049-019-09460-z).
- Schlenker, Philippe (2022): *What It All Means*. Cambridge, MA: MIT Press.
- Schlenker, Philippe, Camille Coye, Shane Steinert-Threlkeld, Nathan Klinedinst & Emmanuel Chemla (2022): Beyond Anthropocentrism in Comparative Cognition: Recentering Animal Linguistics. *Cognitive Science* 46(12). e13220. doi:[10.1111/cogs.13220](https://doi.org/10.1111/cogs.13220).
- Schütze, Carson T. & Jon Sprouse (2013): Judgment data. In Robert J. Podesva & Dvyani Sharma (Hrsg.), *Research Methods in Linguistics*, 27–50. Cambridge University Press. doi:[10.1017/cbo9781139013734.004](https://doi.org/10.1017/cbo9781139013734.004).
- Schwartz, Sandrine (2009): *Stratégies de synchronisation interactionnelle – alternance conversationnelle et rétroaction en cours de discours – chez des locuteurs sourdaveugles pratiquant la Langue des Signes Française tactile (Vol 1)*. University of Paris 8.
- Simcock, Peter (2017): One of society’s most vulnerable groups? A systematically conducted literature review exploring the vulnerability of deafblind people. *Health and Social Care in the Community* 25(3). 813–839. doi:[10.1111/hsc.12317](https://doi.org/10.1111/hsc.12317).
- Taub, Sarah F. (2001): *Language from the body*. Cambridge, UK: Cambridge University Press.
- Templeton, Christopher N., Erick Greene & Kate Davis (2005): Allometry of Alarm Calls: Black-Capped Chickadees Encode Information About Predator Size. *Science* 308(5730). 1934–1937. doi:[10.1126/science.1108841](https://doi.org/10.1126/science.1108841).
- Van Der Mark, Lisa (2023): Deafblind Tactile Signers: The Dynamics of Communication and Space. *Sign Language Studies* 23(4). 500–526. doi:[10.1353/sls.2023.a905537](https://doi.org/10.1353/sls.2023.a905537).
- Volpato, Laura (2023): A preliminary description of haptics in Italian social-haptic communication. *FEAST. Formal and Experimental Advances in Sign language Theory* 5. 210–227. doi:[10.31009/FEAST.i5.17](https://doi.org/10.31009/FEAST.i5.17).
- Volpato, Laura (2024): *Social-haptic communication for the deafblind in Italy: creating, spreading, and analysing Italian haptics*. Ca’ Foscari.

- Warnicke, Camilla, Moa Wahlqvist, Agneta Anderzén-Carlsson & Ann Sofie Sundqvist (2022): Interventions for adults with deafblindness - an integrative review. *BMC Health Services Research*. BioMed Central. 22(1). 1–20. doi:[10.1186/s12913-022-08958-4](https://doi.org/10.1186/s12913-022-08958-4).
- Wilbur, Ronnie B. (2000): Phonological and prosodic layering of nonmanuals in American sign Language. In Karen Emmorey & Harlan Lane (Hrsg.), *The signs of Language Revisited*, 215–244.
- Willoughby, Louisa, Shimako Iwasaki, Meredith Bartlett & Howard Manns (2018): Tactile sign languages Handbook of pragmatics. *Handbook of Pragmatics*, 1–20.
- Willoughby, Louisa, Howard Manns, Iwasaki Shimako & Meredith Bartlett (2014): Misunderstanding and Repair in Tactile Auslan. *Sign Language Studies* 14(4). 419–443. doi:[10.1353/sls.2014.0014](https://doi.org/10.1353/sls.2014.0014).
- Willoughby, Louisa, Howard Manns, Shimako Iwasaki & Meredith Bartlett (2019): Are you trying to be funny? Communicating humour in deafblind conversations. *Discourse Studies* 21(5). 584–602. doi:[10.1177/1461445619846704](https://doi.org/10.1177/1461445619846704).
- Willoughby, Louisa, Howard Manns, Shimako Iwasaki & Meredith Bartlett (2020): From Seeing to Feeling: How Do Deafblind People Adapt Visual Sign Languages? In Keith Allan (Hrsg.), *Dynamics of Language Changes*, 235–252. Singapore: Springer Singapore. doi:[10.1007/978-981-15-6430-7_15](https://doi.org/10.1007/978-981-15-6430-7_15).
- Yoshimura, Hidekane, Takuya Okubo, Jun Shinagawa, Shin-Ya Nishio, Yutaka Takumi & Shin-Ichi Usami (2024): Epidemiology, aetiology and diagnosis of congenital hearing loss via hearing screening of 153 913 newborns. *International Journal of Epidemiology* 53(3). dyae052. doi:[10.1093/ije/dyae052](https://doi.org/10.1093/ije/dyae052).
- Zorzi, Giorgia, Beatrice Giustolisi, Valentina Aristodemo, Carlo Cecchetto, Charlotte Hauser, Josep Quer, Jordina Sánchez Amat & Caterina Donati (2022): On the Reliability of the Notion of Native Signer and Its Risks. *Frontiers in Psychology* 13(March). 1–12. doi:[10.3389/fpsyg.2022.716554](https://doi.org/10.3389/fpsyg.2022.716554).
- Zorzi, Giorgia, Gro Hege Saltnes Urdal & Eli Raanes (2025): Understanding depiction in tactile Norwegian sign language interpreting. *Lingua* 316. 103887. doi:[10.1016/j.lingua.2025.103887](https://doi.org/10.1016/j.lingua.2025.103887).
- Zorzi, Giorgia & Laura Volpato (2025): Cosa sappiamo delle lingue segni tattili nel mondo? Gehalten auf der La comunicazioni e la sordocecità: LIS tattile e comunicazione socio-aplica, Milano, Italy.